

The Kepler Problem and Orbital Perturbation Theory

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1 The Two-Body Problem

1.1 Central force problem

A central force acts along the line joining two particles and depends only on their separation $r = |\vec{x}_1 - \vec{x}_2|$. Such a force is a gradient of a scalar potential $V(r)$:

$$\vec{F}(\vec{x}_1, \vec{x}_2) = \nabla V(r).$$

Newton’s equations for the pair are

$$\frac{d^2}{dt^2} \begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} = \begin{bmatrix} -\nabla V(r) \\ \nabla V(r) \end{bmatrix}. \quad (1)$$

Six second-order ODEs $\Rightarrow$ 12 constants of integration in the general solution.

1.2 Separating centre-of-mass motion

With $M = m_1 + m_2$, $\mu = m_1 m_2 / M$, transform to relative and centre-of-mass coordinates,

$$\begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ m_1/M & m_2/M \end{bmatrix} \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix}, \quad \begin{bmatrix} \vec{x}_1 \\ \vec{x}_2 \end{bmatrix} = \begin{bmatrix} m_2/M & 1 \\ -m_1/M & 1 \end{bmatrix} \begin{bmatrix} \vec{r} \\ \vec{X} \end{bmatrix}.$$

Substituting into (1) and left-multiplying by $\begin{bmatrix} m_2/M & -m_1/M \\ 1 & 1 \end{bmatrix}$ diagonalizes the mass matrix and decouples the two sectors:

$$\mu \ddot{\vec{r}} = -\nabla V(r), \quad (2)$$

$$M \ddot{\vec{X}} = 0. \quad (3)$$

Equation (3) integrates trivially: $\vec{X}(t) = (\vec{P}/M)t + \vec{X}_0$, with $\vec{P}, \vec{X}_0$ six constants fixing the inertial frame in which the barycentre is at rest at the origin. Everything of dynamical interest is in the *relative* coordinate $\vec{r}$, governed by (2) — a one-body central-force problem with reduced mass μ .

1.3 Conserved quantities

Angular momentum. Define $\vec{L} = \vec{r} \times \vec{p} = \mu \vec{r} \times \dot{\vec{r}}$. Then, because we are considering central force $\ddot{\vec{r}} \propto \hat{r}$ and hence,

$$\dot{\vec{L}} = \mu [\dot{\vec{r}} \times \dot{\vec{r}} + \vec{r} \times \ddot{\vec{r}}] = 0.$$

$\vec{L}$ supplies three constants of motion: its magnitude L , and two angles fixing its direction — the **inclination** ι and **longitude of ascending node** Ω (Euler angles of the orbital plane relative to a reference plane). Conservation of $\vec{L}$ confines the motion to the plane $\perp \vec{L}$.

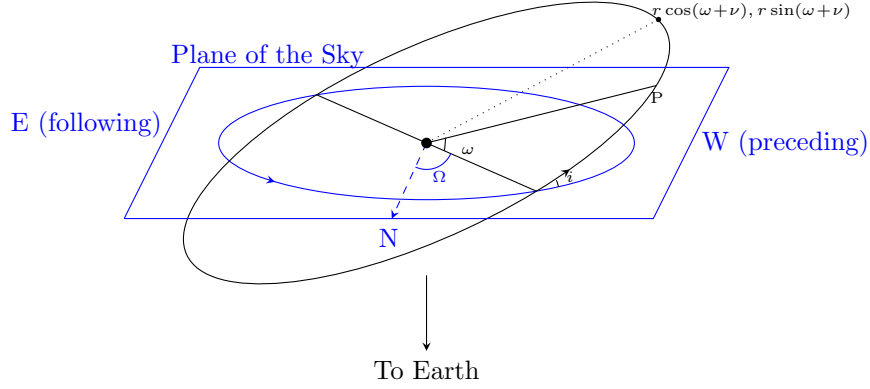


Figure 1: In this North becomes the local X axis and West becomes the Y axis of the sky plane.

Energy. $E = K + V$ with $K = \frac{1}{2}\mu\dot{r}^2$ (the centre-of-mass kinetic term is a constant and drops out). Then

$$\dot{E} = \mu\dot{\vec{r}} \cdot \ddot{\vec{r}} + \dot{V} = \mu\dot{\vec{r}} \cdot \ddot{\vec{r}} + \nabla V \cdot \dot{\vec{r}} = (\mu\ddot{\vec{r}} + \nabla V) \cdot \dot{\vec{r}} = 0$$

by (2). So E is conserved for *any* central potential $V(r)$, not just gravity.

2 Coordinate Systems and Orbital Elements

Fix an equatorial (reference) frame with $\hat{\Upsilon}$ (vernal equinox / first point of Aries) along X and celestial north $\hat{N}$ along Z ; $\hat{Y}$ completes the right-handed triad. For a source at $(\text{RA}, \text{DEC}) = (\alpha, \delta)$ the line of sight is

$$\hat{R} = \cos \delta \cos \alpha \hat{\Upsilon} + \cos \delta \sin \alpha \hat{Y} + \sin \delta \hat{N},$$

with in-sky-plane axes (pointing along increasing RA, DEC)

$$\hat{\alpha} = \frac{\partial \hat{R}}{\partial \alpha} = -\sin \alpha \hat{\Upsilon} + \cos \alpha \hat{Y}, \quad \hat{\delta} = \frac{\partial \hat{R}}{\partial \delta} = -\sin \delta \cos \alpha \hat{\Upsilon} - \sin \delta \sin \alpha \hat{Y} + \cos \delta \hat{N}.$$

The orbital plane has its own frame: origin at the barycentre, X -axis along the **line of nodes** (through the ascending node and the descending node), Z -axis along $\vec{L}$. Denote the unit vector towards the ascending node by $\hat{\delta}$. Then

$$\begin{pmatrix} \hat{\delta} \\ \hat{\gamma} \\ \hat{L} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \iota & \sin \iota \\ 0 & -\sin \iota & \cos \iota \end{pmatrix} \begin{pmatrix} \cos \Omega & \sin \Omega & 0 \\ -\sin \Omega & \cos \Omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \hat{\alpha} \\ \hat{\delta} \\ \hat{R} \end{pmatrix}. \quad (4)$$

Hence,

$$\hat{\Omega} = \cos \Omega \hat{\alpha} + \sin \Omega \hat{\delta}, \quad (5)$$

$$\hat{y} = -\sin \Omega \cos \iota \hat{\alpha} + \cos \Omega \cos \iota \hat{\delta} + \sin \iota \hat{R}, \quad (6)$$

$$\hat{L} = \sin \Omega \sin \iota \hat{\alpha} - \cos \Omega \sin \iota \hat{\delta} + \cos \iota \hat{R}, \quad (7)$$

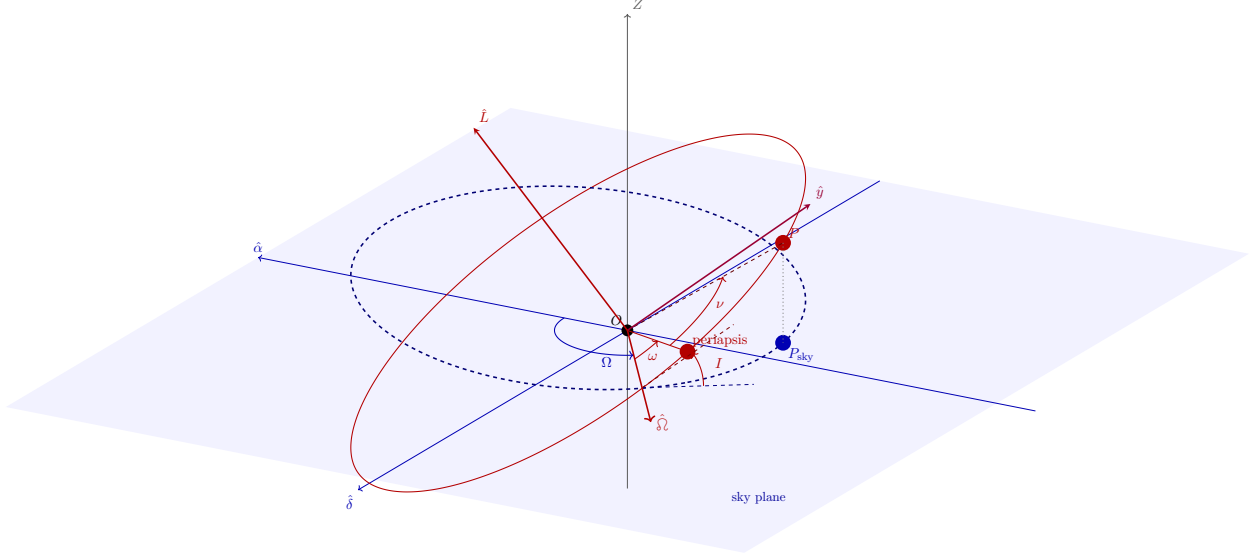


Figure 2: Orientation of an orbit relative to sky plane. Ω (longitude of ascending node) is measured from the reference direction α to the ascending node along the sky plane; ω (argument of periapsis) is measured from the ascending node to perigee *within* the orbital plane; ι (inclination) is the dihedral angle between the two planes, read off at the node; ν (true anomaly) is measured from perigee to the heavenly body's current position. This is exactly (Ω, ι, ω) of Eqs. (5)–(6) describe.

so that $\iota = \arccos(\hat{L} \cdot \hat{R})$ is the angle between orbital and reference planes, and $(\hat{\Omega}, \hat{y}, \hat{L})$ is a right-handed orthonormal triad spanning the orbital plane plus its normal. In this plane use polar coordinates (r, ϕ) measured from $\hat{\Omega}$:

$$\hat{r} = \cos \phi \hat{\Omega} + \sin \phi \hat{y}, \quad \hat{\phi} = -\sin \phi \hat{\Omega} + \cos \phi \hat{y}. \quad (8)$$

The projected orbit in the sky plane : If the object in its orbital plane as the coordinate $(r \cos(\omega + \nu), r \sin(\omega + \nu))$, the corresponding coordinate in the sky plane could be found as:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_{\text{Sky Plane}} = \begin{pmatrix} \cos \Omega & -\sin \Omega & 0 \\ \sin \Omega & \cos \Omega & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos i & -\sin i \\ 0 & \sin i & \cos i \end{pmatrix} \begin{pmatrix} r \cos(\omega + \nu) \\ r \sin(\omega + \nu) \\ 0 \end{pmatrix}_{\text{Orbital Plane}}. \quad (9)$$

Then,

$$X = r (\cos(\omega + \nu) \cos \Omega - \sin(\omega + \nu) \sin \Omega \cos i), \quad (10)$$

$$Y = r (\cos(\omega + \nu) \sin \Omega + \sin(\omega + \nu) \cos \Omega \cos i). \quad (11)$$

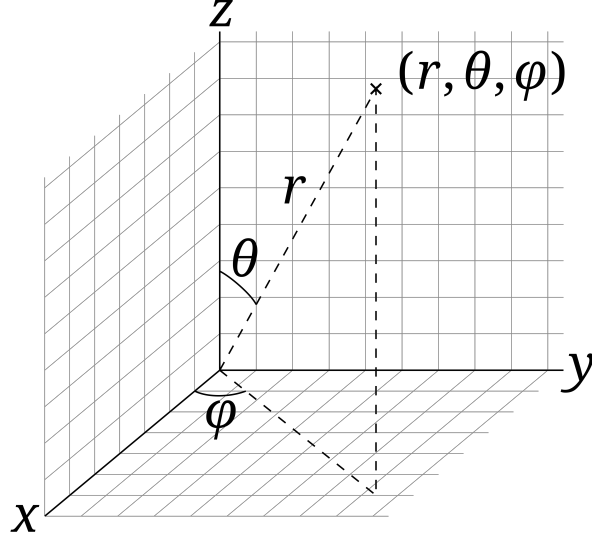


Figure 3: Schematic description of spherical polar coordinates.

One can set $\Omega = 0$ by another set of basis in the sky plane and then the analysis simplifies immensely

$$X = r \cos(\omega + \nu), \quad (12)$$

$$Y = r \sin(\omega + \nu) \cos i. \quad (13)$$

We find that the semi minor axis shrinks by the factor of $\cos i$.

The six orbital elements. As the orbit solution unfolds below, six constants emerge in total: Ω, ι (orientation of the plane, from $\vec{L}$), a, e (size and shape, from E, L), ω (orientation of the ellipse *within* the plane, the **argument of periapsis**), and T_0 (time of periapsis passage, fixing where on the ellipse the body is at a given time). Figure 2 shows all of these together.

3 Planar Motion and Kepler's Second Law

Areal velocity. The area swept by $\vec{r}$ in time dt is $dA = \int_0^{2\pi} \int_0^{r(\phi)} r dr d\phi = \frac{1}{2} \int_0^{2\pi} r^2(\phi) d\phi$, so

$$\frac{dA}{dt} = \frac{r^2 \dot{\phi}}{2} = \frac{L}{2\mu} = \text{const.} \quad (14)$$

This is **Kepler's second law**: equal areas in equal times. It follows purely from $\vec{L}$ being conserved, i.e. from the force being central — no dependence on the specific form of $V(r)$.

Equations of motion in (r, ϕ) . With $Z \parallel \vec{L}$ and $X \parallel \hat{\delta}\hat{\Omega}$, $\vec{r} = r\hat{r}$, and

$$\dot{\vec{r}} = \dot{r}\hat{r} + r\dot{\hat{r}} = \dot{r}\hat{r} + r\dot{\phi}\hat{\phi}, \quad \ddot{\vec{r}} = (\ddot{r} - r\dot{\phi}^2)\hat{r} + (2\dot{r}\dot{\phi} + r\ddot{\phi})\hat{\phi}.$$

where we used $\nabla_\mu \mathbf{e}_\nu = \Gamma_{\mu\nu}^\alpha \mathbf{e}_\alpha$ which gives $\frac{d\hat{r}}{dt} = \frac{\partial \hat{r}}{\partial \phi} \dot{\phi} = \dot{\phi} \Gamma_{\phi r}^\mu \mathbf{e}_\mu = \dot{\phi} \frac{\mathbf{e}_\phi}{r} = \dot{\phi} \hat{\phi}$. Equation (2) splits into radial and tangential parts:

$$\mu(\ddot{r} - r\dot{\phi}^2) = -\nabla V(r), \quad (15)$$

$$\mu(2\dot{r}\dot{\phi} + r\ddot{\phi}) = 0 \implies \frac{d(r^2\dot{\phi})}{dt} = 0. \quad (16)$$

This was supposed to be a coupled differential equation. But because of equation (16) is just angular momentum conservation, i.e. Kepler's second law again, since $\vec{L} = \mu \vec{r} \times \dot{\vec{r}} = \mu r^2 \dot{\phi} \hat{L}$. So

$$\dot{\phi} = \frac{L}{\mu r^2}, \quad (17)$$

and (15) becomes, eliminating $\dot{\phi}$,

$$\mu \ddot{r} - \frac{L^2}{\mu r^3} = -\nabla V(r). \quad (18)$$

4 The Orbit Equation

To get the orbit *shape* $r(\phi)$ we change independent variable from t to ϕ using (17): $\frac{dr}{dt} = \dot{\phi} \frac{dr}{d\phi} = \frac{L}{\mu r^2} \frac{dr}{d\phi}$, and

$$\frac{d^2 r}{dt^2} = \frac{L}{\mu r^2} \frac{d}{d\phi} \left[\frac{L}{\mu r^2} \frac{dr}{d\phi} \right] = \frac{L^2}{\mu^2 r^4} \left[\frac{d^2 r}{d\phi^2} - \frac{2}{r} \left(\frac{dr}{d\phi} \right)^2 \right].$$

Substituting into (18) and dividing by $L^2/(\mu r^4)$:

$$\frac{1}{r^4} \left[\frac{d^2 r}{d\phi^2} - \frac{2}{r} \left(\frac{dr}{d\phi} \right)^2 \right] - \frac{1}{r^3} = -\frac{\mu}{L^2} \nabla V(r).$$

The standard trick is the **Binet substitution** $r = 1/\rho$: then $dr/d\phi = -\rho^{-2} d\rho/d\phi$ and $d^2 r/d\phi^2 = -\rho^{-2} d^2 \rho/d\phi^2 + 2\rho^{-3} (d\rho/d\phi)^2$. Substituting and simplifying (the $(d\rho/d\phi)^2$ terms cancel), one obtains the **orbit equation**:

$$\boxed{\frac{d^2 \rho}{d\phi^2} + \rho = \frac{\mu}{L^2} \frac{1}{\rho^2} \nabla V\left(\frac{1}{\rho}\right)}. \quad (19)$$

This is a purely geometric ODE for the orbit's shape, valid for any central potential.

5 Keplerian Orbits

For Newtonian gravity, $V(r) = -GM\mu/r$, so $\nabla V(r) = GM\mu\rho^2 \hat{r}$. Equation (19) becomes

$$\frac{d^2 \rho}{d\phi^2} + \rho = \frac{GM\mu^2}{L^2}.$$

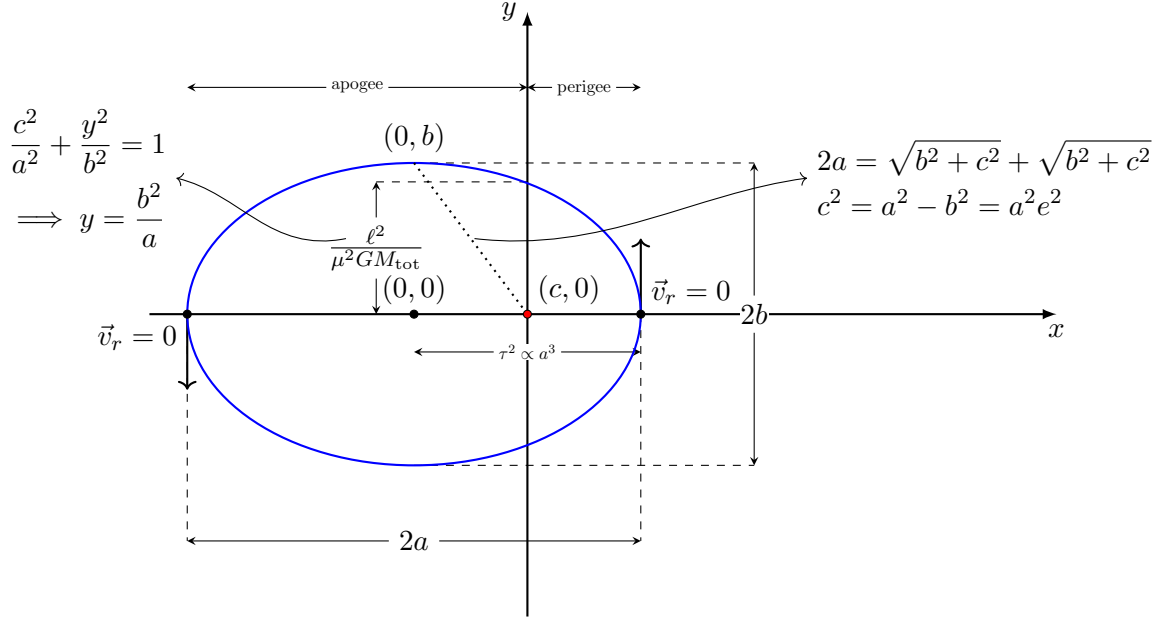
Shifting $\tilde{\rho} = \rho - GM\mu^2/L^2$ turns this into the simple-harmonic-oscillator equation $\ddot{\tilde{\rho}} + \tilde{\rho} = 0$ (derivatives in ϕ), with solution $\tilde{\rho} = A \cos(\phi - \omega)$ for constants A, ω . In terms of $r = 1/\rho$:

$$r(\phi) = \left(\frac{GM\mu^2}{L^2} + A \cos(\phi - \omega) \right)^{-1} = \frac{L^2/(GM\mu^2)}{1 + \frac{L^2 A}{GM\mu^2} \cos(\phi - \omega)}.$$

Defining the **semi-latus rectum** $\lambda \equiv L^2/(GM\mu^2)$ as the value of r when $\phi = \omega + 90^\circ$ and **eccentricity** $e \equiv \lambda A$,

$$\boxed{r(\phi) = \frac{\lambda}{1 + e \cos(\phi - \omega)}}. \quad (20)$$

This is the polar equation of a conic section with one focus at the origin, semi-latus rectum λ , eccentricity e , and major axis rotated by ω (the **argument of periapsis** — periapsis occurs at $\phi = \omega$).



Two more shape parameters follow:

$$a = \frac{\lambda}{1 - e^2} \text{ (semi-major axis),} \quad b = a\sqrt{1 - e^2} \text{ (semi-minor axis).}$$

One can check this directly from the equation of the ellipse,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad (21)$$

by writing the polar coordinates about the focus as

$$x = ae + r(\phi) \cos \phi, \quad (22)$$

$$y = r(\phi) \sin \phi. \quad (23)$$

Substitution gives

$$\frac{(ae + r \cos \phi)^2}{a^2} + \frac{r^2 \sin^2 \phi}{a^2(1 - e^2)} = 1. \quad (24)$$

Solving for r then yields

$$r(\phi) = \frac{a(1 - e^2)}{1 + e \cos \phi}. \quad (25)$$

Eccentricity	Conic section
$e = 0$	circle
$0 < e < 1$	ellipse ($E < 0$, bound)
$e = 1$	parabola ($E = 0$)
$e > 1$	hyperbola ($E > 0$, unbound)

Bound, closed elliptical orbits ($E < 0$) constitute **Kepler's first law**. In the original (unreduced) frame,

$$\vec{x}_1 = \frac{m_2}{M} \vec{r}, \quad \vec{x}_2 = -\frac{m_1}{M} \vec{r}.$$

6 Energy and Angular Momentum in Terms of a, e

6.1 Energy

$$E = \frac{1}{2} \mu \dot{r}^2 - \frac{GM\mu}{r} = \frac{1}{2} \mu (\dot{r}^2 + r^2 \dot{\phi}^2) - \frac{GM\mu}{r} = \frac{1}{2} \mu \dot{r}^2 + \frac{L^2}{2\mu r^2} - \frac{GM\mu}{r}.$$

Since E is conserved, evaluate at periapsis ($\phi = \omega$), where $r = a - ae = \lambda/(1+e)$ and $\dot{r} = 0$ (turning point of r):

$$E = 0 + \frac{L^2(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} = \frac{GM\mu^2\lambda(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} = -\frac{GM\mu}{2\lambda}(1-e^2),$$

using $L^2 = GM\mu^2\lambda$ (below). With $a = \lambda/(1-e^2)$,

$$\boxed{E = -\frac{GM\mu}{2a}}. \quad (26)$$

Since $\lambda > 0$ always, the sign of E tracks e : bound orbits ($E < 0$) have $e < 1$, $a > 0$; unbound orbits ($E \geq 0$) have $e \geq 1$; hyperbolic orbits formally have $a < 0$.

6.2 Angular momentum

Both at apoapsis ($r = a + ae = \frac{\lambda}{1-e}$) and periapsis ($r = a - ae = \frac{\lambda}{1+e}$), we have $\dot{r} = 0$,

$$\frac{L^2(1+e)^2}{2\mu\lambda^2} - \frac{GM\mu(1+e)}{\lambda} = \frac{L^2(1-e)^2}{2\mu\lambda^2} - \frac{GM\mu(1-e)}{\lambda} \quad (27)$$

Solving for L :

$$L = \sqrt{GM\mu^2\lambda}, \quad \text{i.e.} \quad L^2 = GM\mu^2\lambda(1-e^2), \quad (28)$$

and, using (7),

$$\vec{L} = \sqrt{GM\mu^2\lambda} (-\sin\Omega \sin\iota \hat{\alpha} + \cos\Omega \sin\iota \hat{\delta} + \cos\iota \hat{R}).$$

6.3 Instantaneous kinetic energy

From $E = K + V$: $K = \frac{1}{2} \mu \dot{r}^2 = E - V = GM\mu \left(\frac{1}{r} - \frac{1}{2a} \right)$ — the *vis-viva* equation. Splitting between the two bodies via $\vec{x}_i$: $K_1 = \frac{1}{2} m_1 v_1^2 = \frac{m_2}{M} K$, $K_2 = \frac{m_1}{M} K$.

Inverting for a, e . Equations (26)–(28) give the orbital elements directly from the (conserved) energy and angular momentum:

$$a = -\frac{GM\mu}{2E}, \quad e = \sqrt{1 + \frac{2EL^2}{(GM)^2\mu^3}}. \quad (29)$$

7 True and Eccentric Anomaly; Kepler's Equation

We now want $r(t)$ (and hence $\phi(t)$), not just $r(\phi)$. From the vis-viva relation,

$$\dot{r} = \sqrt{\frac{2}{\mu} \left(E - \frac{L^2}{2\mu r^2} + \frac{GM\mu}{r} \right)} = \sqrt{GM} \frac{1}{r} \left(2r - \frac{r^2}{a} - \lambda \right)^{1/2},$$

using (26)–(28) to trade E, L for a, λ . Then

$$t - t_0 = \frac{1}{\sqrt{GM}} \int_{r_0}^r \frac{r' dr'}{(2r' - r'^2/a - \lambda)^{1/2}}.$$

7.1 Elliptical orbits: eccentric anomaly

For $a > 0$, substitute

$$r = a(1 - e \cos u), \quad (30)$$

where u is the **eccentric anomaly** (this choice guarantees $r > 0$ throughout, since $-1 \leq \cos u \leq 1$). Then $dr = ae \sin u du$, and the bracket under the square root simplifies (using $\lambda = a(1 - e^2)$) to $ae^2 \sin^2 u$, so the square root is simply $\sqrt{a} e |\sin u|$. The integral collapses to

$$t - t_0 = \frac{a^{3/2}}{\sqrt{GM}} \int_0^u (1 - e \cos u') du' = \frac{a^{3/2}}{\sqrt{GM}} [u - e \sin u],$$

where t_0 (**epoch**) is the time at which $u = 0$ — the sixth and final constant of integration, T_0 .

Relating true anomaly v and eccentric anomaly u : Define the **true anomaly** $v \equiv \phi - \omega$ (so $r = \lambda/(1 + e \cos v)$ from (20)). Equating this with (30):

$$\frac{1 - e \cos u}{1 - e^2} = \frac{1}{1 + e \cos v} \implies \cos v = \frac{\cos u - e}{1 - e \cos u}. \quad (65a)$$

Consistently,

$$\sin v = \frac{\sqrt{1 - e^2} \sin u}{1 - e \cos u}, \quad \tan v = \frac{\sqrt{1 - e^2} \sin u}{\cos u - e}, \quad \tan \frac{v}{2} = \sqrt{\frac{1 + e}{1 - e}} \tan \frac{u}{2}. \quad (31)$$

At $u = 0$ and $u = \pi$, $v = u$ — periapsis and apoapsis coincide for the two anomalies, as they must by symmetry.

7.2 Hyperbolic trajectories

For $a < 0$ (and $e > 1$), the analogous substitution is $r = a(1 - e \cosh u)$ (using $\cosh u \geq 1$ keeps $r > 0$ since $a < 0$). Carrying through the same integral gives the **hyperbolic** anomaly relation

$$t - t_0 = \frac{(-a)^{3/2}}{\sqrt{GM}} [e \sinh u - u],$$

with true-anomaly relations

$$\cos v = \frac{e - \cosh u}{e \cosh u - 1}, \quad \sin v = \frac{\sqrt{e^2 - 1} \sinh u}{e \cosh u - 1}, \quad \tan \frac{v}{2} = \sqrt{\frac{e + 1}{e - 1}} \tanh \frac{u}{2}. \quad (32)$$

8 Kepler's Third Law

For one complete orbit, u advances by 2π while t advances by the orbital period $2\pi/n$ ($n \equiv$ mean motion):

$$\frac{2\pi}{n} = \frac{a^{3/2}}{\sqrt{GM}} [2\pi - e \sin(2\pi)] = \frac{a^{3/2}}{\sqrt{GM}} \cdot 2\pi \implies \boxed{n = \sqrt{\frac{GM}{a^3}}, \quad n^2 a^3 = GM.} \quad (33)$$

9 Kepler's Equation

Substituting $n = \sqrt{GM/a^3}$ into $t - t_0 = (a^{3/2}/\sqrt{GM})(u - e \sin u)$ and defining the **mean anomaly** $\ell \equiv n(t - t_0)$:

$$\boxed{\ell = u - e \sin u} \quad (\text{elliptical Kepler's equation}). \quad (34)$$

This transcendental equation must be solved numerically (or by series, §12) for $u(\ell)$, i.e. $u(t)$. The hyperbolic analogue (with $n \equiv GM/(-a)^{3/2}$) is

$$\ell = e \sinh u - u. \quad (35)$$

10 Keplerian Parametrization

10.1 Elliptical orbits

Collecting results:

$$r = a(1 - e \cos u), \quad \phi = v + \omega, \quad (36)$$

$$\tan \frac{v}{2} = \sqrt{\frac{1-e}{1+e}} \tan \frac{u}{2}, \quad \ell = u - e \sin u. \quad (37)$$

Differentiating Kepler's equation (34) in t : $n = \dot{u}(1 - e \cos u)$, so

$$\dot{u} = \frac{n}{1 - e \cos u}. \quad (38)$$

Differentiating $r = a(1 - e \cos u)$:

$$\dot{r} = ae \sin u \dot{u} = \frac{ane \sin u}{1 - e \cos u}. \quad (39)$$

Differentiating the v - u relation and using (31), one finds $dv/du = \sqrt{1-e^2}/(1 - e \cos u)$, so

$$\dot{\phi} = \dot{v} = \frac{dv}{du} \dot{u} = \frac{n\sqrt{1-e^2}}{(1 - e \cos u)^2}. \quad (40)$$

10.2 Hyperbolic trajectories

Analogously, with $r = -a(e \cosh u - 1)$, $\phi = v + \omega$, $\tan(v/2) = \sqrt{(e+1)/(e-1)} \tanh(u/2)$, $\ell = e \sinh u - u$:

$$\dot{u} = \frac{n}{e \cosh u - 1}, \quad \dot{r} = -an \frac{e \sinh u}{e \cosh u - 1}, \quad \dot{\phi} = \dot{v} = \frac{n\sqrt{e^2-1}}{(e \cosh u - 1)^2}. \quad (41)$$

11 Hyperbolic-Trajectory Parameters

For unbound ($e > 1$) encounters, four derived quantities are standard:

Hyperbolic excess velocity v_∞ : from $\frac{1}{2}\mu\dot{r}_\infty^2 = -GM\mu/(2a)$ (recall $a < 0$),

$$|\dot{r}_\infty| = \sqrt{-\frac{GM}{a}}. \quad (42)$$

Scattering angle θ_∞ : half the angle between the two asymptotes,

$$\cos \theta_\infty = - \lim_{u \rightarrow \infty} \frac{e - \cosh u}{e \cosh u - 1} = \frac{1}{e}. \quad (43)$$

Periapsis distance:

$$r_p = \min_v \frac{a(1 - e^2)}{1 + e \cos v} = -a(e - 1). \quad (44)$$

Semi-minor axis / impact parameter: the semi-minor axis $b = -a\sqrt{e^2 - 1}$ satisfies $-ab = \lambda$. The impact parameter b' (miss distance of the unperturbed asymptote) is

$$b' = (-a + r_p) \sin \theta_\infty = -ae \sin \theta_\infty = -a\sqrt{e^2 - 1} = b,$$

i.e. impact parameter and semi-minor axis coincide.

Flight-path angle (angle between $\dot{\vec{r}}$ and the direction $\perp \hat{r}$ in-plane):

$$\sin \varphi = \frac{\dot{\vec{r}} \cdot \hat{r}}{|\dot{\vec{r}}|} = \frac{\dot{r}}{\sqrt{\dot{r}^2 + r^2 \dot{\phi}^2}}. \quad (45)$$

12 Fourier Solution of Kepler's Equation

Kepler's equation $u - \ell = e \sin u$ can be solved as a Fourier series in ℓ :

$$e \sin u = \sum_{s=1}^{\infty} A_s \sin(s\ell).$$

Using $A_s = \frac{2}{\pi} \int_0^\pi (u - \ell) \sin(s\ell) d\ell$, integrating by parts (the boundary term vanishes since $u = \ell$ at $\ell = 0, \pi$), and substituting $d\ell = (1 - e \cos u) du$ (from Kepler's equation) turns the ℓ -integral into a u -integral:

$$A_s = \frac{2}{s\pi} \int_0^\pi \cos(su - se \sin u) du = \frac{2}{s} J_s(se),$$

recognizing the integral representation of the Bessel function of the first kind, $J_s(x) = \frac{1}{\pi} \int_0^\pi \cos(s\tau - x \sin \tau) d\tau$. So

$$u(\ell) = \ell + \sum_{s=1}^{\infty} \frac{2}{s} J_s(se) \sin(s\ell). \quad (46)$$

This is the classical Bessel-function series solution to Kepler's equation.

13 The Laplace–Runge–Lenz Vector

Define $\vec{A} \equiv \vec{p} \times \vec{L} - GM\mu^2 \hat{r}$ (with $\vec{p} = \mu \dot{\vec{r}}$). For the pure Kepler problem,

$$\dot{\vec{A}} = \dot{\vec{p}} \times \vec{L} - GM\mu^2 \dot{\hat{r}} = -\frac{GM\mu}{r^2} (\hat{r} \times \vec{L}) - GM\mu^2 \dot{\phi} \hat{\phi}.$$

Using $\vec{L} = L\hat{L}$, planar motion identities $\hat{r} \times \hat{L} = -\hat{\phi}$, and $\dot{\phi} = L/(\mu r^2)$ from (17):

$$\dot{\vec{A}} = -GM\mu \left(\frac{L}{r^2} \hat{\phi} - \mu \dot{\phi} \hat{\phi} \right) = 0,$$

so $\vec{A}$ is conserved for the unperturbed problem — it is a *fourth*, non-obvious constant of the Kepler problem (beyond $E, \vec{L}$), reflecting the extra "hidden" $SO(4)$ symmetry of the $1/r$ potential.

Direction and magnitude of $\vec{A}$. From the definition it is clear that $\vec{A}$ is planar. Therefore, we can decompose it as $\vec{A} = A_r \hat{r} + A_\phi \hat{\phi}$.

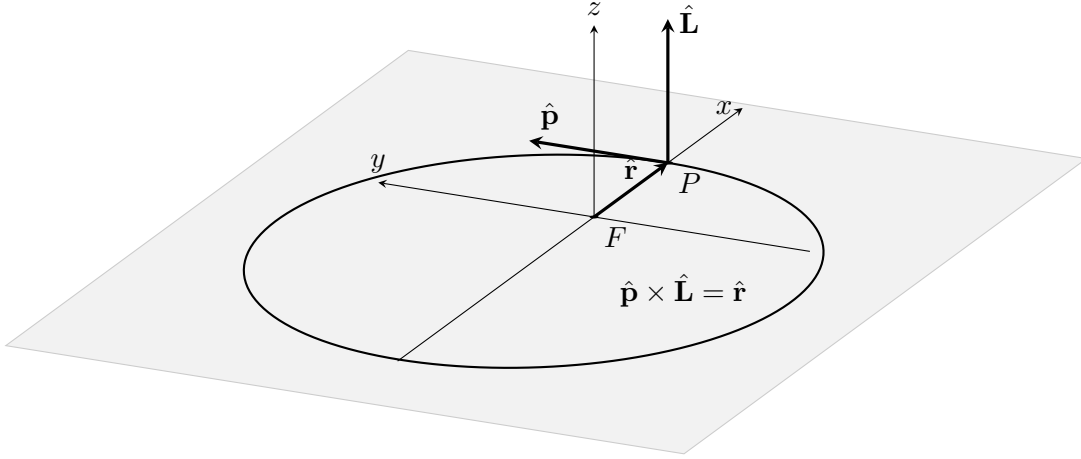


Figure 4: The Runge Lenz vector points radially outward at perigee. Since it is conserved, the direction and magnitude stays fixed during the entire evolution.

At periapsis the velocity is purely tangential ($\vec{p} \perp \hat{r}$), so $\hat{r} \times \vec{A}|_{\text{peri}} = \hat{r} \times (\vec{p} \times \vec{L})|_{\text{peri}} = [\vec{p}(\hat{r} \cdot \vec{L}) - \vec{L}(\hat{r} \cdot \vec{p})]|_{\text{peri}} = 0$ (both dot products vanish there). Also $\hat{r} \cdot \vec{A}|_{\text{peri}} = A$ (from the relation above with $v = 0$ at periapsis). Together these mean $\hat{A} = \hat{r}|_{\text{peri}}$: $\vec{A}$ points from the focus *towards periapsis*.

Since $\vec{A}$ points radially outwards at the Periapsis. We find the magnitude by dotting $\vec{A}$ into $\vec{r}$:

$$\vec{A} \cdot \vec{r} = (\vec{p} \times \vec{L}) \cdot \vec{r} - GM\mu^2 r \tag{47}$$

$$= p_i L_j \epsilon_{ijk} r_k - GM\mu^2 r = p_i L_j \epsilon_{kij} r_k - GM\mu^2 r \tag{48}$$

$$= \underbrace{(\vec{r} \times \vec{p}) \cdot \vec{L}}_{\vec{L}} - GM\mu^2 r = L^2 - GM\mu^2 r, \tag{49}$$

i.e., writing $\vec{A} \cdot \vec{r} = Ar \cos v$ (with v the angle between $\vec{A}$ and $\vec{r}$), using

$$r = \frac{L^2/(GM\mu^2)}{1 + e \cos v} \implies GM\mu^2 r = \frac{L^2}{1 + e \cos v}.$$

We find:

$$Ar \cos v = L^2 - GM\mu^2 r, \quad (50)$$

$$A \frac{L^2/GM\mu^2}{1 + e \cos v} \cos v = L^2 \left(1 - \frac{1}{1 + e \cos v} \right) = \frac{L^2 e \cos v}{1 + e \cos v} \quad (51)$$

Hence

$$\boxed{A = GM\mu^2 e.} \quad (52)$$

Using (8):

$$\hat{A} = \cos \omega \hat{\mathcal{G}} + \sin \omega \hat{y}. \quad (53)$$

Equivalently, the **eccentricity vector** $\vec{e} \equiv \vec{A}/(GM\mu^2) = e\hat{A}$ is a conserved vector encoding (e, ω) together. Since e, Ω, ι are already determined by $(E, \vec{L})$, the only *new* information in $\vec{A}$ is ω — the in-plane orientation of the ellipse.

14 Perturbed Orbits

Now add a perturbing force $\vec{F}$ (not necessarily central) to the Kepler problem:

$$\mu \ddot{\vec{r}} + \frac{GM\mu}{r^2} \hat{r} = \vec{F}. \quad (54)$$

14.1 The osculating-orbit ansatz (variation of parameters)

Before writing down rates for $a, e, \iota, \Omega, \omega$, it is worth being explicit about *why* this is a sensible thing to do at all — this justification was skipped in the reference notes but is worth reconstructing, since it is what turns “ a ” from a constant of the unperturbed problem into a well-defined, instantaneously-meaningful function $a(t)$ of the perturbed one.

For the unperturbed ($\vec{F} = 0$) problem, the solution is fully determined by six constants Q_i ($i = 1, \dots, 6$) — take these to be $(a, e, \iota, \Omega, \omega, T_0)$ — through fixed functional forms

$$\vec{r} = \vec{r}(Q_1, \dots, Q_6, t), \quad \dot{\vec{r}} = \dot{\vec{r}}(Q_1, \dots, Q_6, t)$$

(these are exactly the formulas of §10 translated into Cartesian orbital-plane components via (8)). With $\vec{F} \neq 0$ the true trajectory is no longer a fixed ellipse, but at every instant it is tangent to *some* ellipse. The **osculating-orbit ansatz** keeps the same functional forms but promotes $Q_i \rightarrow Q_i(t)$, and *defines* the instantaneous ellipse by requiring that the true velocity still be given by the same function $\dot{\vec{r}}(Q_i, t)$ that described the unperturbed problem — i.e., that position and velocity are simultaneously matched to some instantaneous Kepler orbit, not just position alone.

Concretely: differentiating $\vec{r}(Q_i(t), t)$,

$$\frac{d\vec{r}}{dt} = \left. \frac{\partial \vec{r}}{\partial t} \right|_{Q_i} + \sum_{i=1}^6 \frac{\partial \vec{r}}{\partial Q_i} \frac{dQ_i}{dt}. \quad (55)$$

By construction $\partial \vec{r} / \partial t|_{Q_i} = \dot{\vec{r}}(Q_i, t)$. Requiring the true velocity to equal this same function forces

$$\sum_{i=1}^6 \frac{\partial \vec{r}}{\partial Q_i} \frac{dQ_i}{dt} = 0. \quad (56)$$

This is a vector equation (3 scalar constraints). It is a *gauge choice*, not extra physics — it is exactly what makes the instantaneous ellipse ”osculating” (tangent in both position and velocity), and it is always solvable for the $\dot{Q}_i$ alongside the equation below, so no generality is lost.

Differentiating $\vec{r}(Q_i(t), t)$ similarly:

$$\frac{d\vec{r}}{dt} = \left. \frac{\partial \vec{r}}{\partial t} \right|_{Q_i} + \sum_{i=1}^6 \frac{\partial \vec{r}}{\partial Q_i} \frac{dQ_i}{dt}.$$

Since $\vec{r}(Q_i, t)$ solves the *unperturbed* problem for fixed Q_i , $\partial \vec{r} / \partial t|_{Q_i} = -GM\hat{r}/r^2$. The left side is the true acceleration, given by (54). Equating and cancelling the common $-GM\hat{r}/r^2$:

$$\sum_{i=1}^6 \frac{\partial \vec{r}}{\partial Q_i} \frac{dQ_i}{dt} = \frac{\vec{F}}{\mu}. \quad (57)$$

Another vector equation, 3 more scalar constraints. Together, (56) and (57) give $3 + 3 = 6$ scalar equations for the 6 unknowns $\dot{Q}_i$ — an exactly determined linear system (given the Jacobian relating $(\vec{r}, \dot{\vec{r}})$ to (Q_i) is non-singular, true away from degenerate orbits such as $e = 0$ or $\iota = 0$). Solving this system for the $\dot{Q}_i$ is precisely what produces the Lagrange/Gauss planetary equations derived below — e.g. Eqs. (68), (69), (67), (59). The derivations of $\dot{E}, \dot{\vec{L}}, \dot{\vec{A}}$ that follow are exact identities for the true trajectory and do not by themselves need this ansatz; what the ansatz supplies is the licence to *read* $a(t) = -GM\mu/2E(t)$, etc., as the semi-major axis of a genuine instantaneous ellipse rather than just a convenient bookkeeping variable.

14.2 Rates of the fundamental constants

Repeating the conservation-law derivations of §1.3 with the extra force:

$$\dot{E} = \dot{\vec{r}} \cdot \vec{F}, \quad (58)$$

$$\dot{\vec{L}} = \vec{r} \times \vec{F}, \quad (59)$$

$$\dot{\vec{A}} = \vec{F} \times \vec{L} + \mu \dot{\vec{r}} \times (\vec{r} \times \vec{F}). \quad (60)$$

For (58): the Kepler-force terms cancel exactly as in §1.3 (that cancellation used nothing but $\vec{F} = 0$ for those terms), leaving only the work done by $\vec{F}$. For (59): the Kepler-force contribution $\vec{r} \times (-GM\mu\hat{r}/r^2)$ vanishes identically since $\vec{r} \parallel \hat{r}$, leaving only the torque from $\vec{F}$. For $\dot{\vec{A}}$: since $\vec{A}$ is exactly conserved when $\vec{F} = 0$, every term *not* containing $\vec{F}$ in the full expansion must sum to zero (this is the same cancellation verified explicitly for the α/r^3 example in §16), leaving

$$\dot{\vec{A}} = \vec{F} \times \vec{L} + \mu \dot{\vec{r}} \times (\vec{r} \times \vec{F}).$$

Expand the double cross product $(\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a} \cdot \vec{c}) - \vec{c}(\vec{a} \cdot \vec{b}))$:

$$\mu \dot{\vec{r}} \times (\vec{r} \times \vec{F}) = \mu \vec{r}(\dot{\vec{r}} \cdot \vec{F}) - \mu \vec{F}(\dot{\vec{r}} \cdot \vec{r}) = \mu \vec{r}(\dot{\vec{r}} \cdot \vec{F}) - \mu r \dot{\vec{r}} \cdot \vec{F}.$$

Writing $\vec{r} = r\hat{r}$ and factoring $\mu r \dot{\vec{r}}$:

$$\dot{\vec{A}} = \vec{F} \times \vec{L} + \mu r \dot{\vec{r}} \left[\frac{(\dot{\vec{r}} \cdot \vec{F})}{\dot{r}} \hat{r} - \vec{F} \right].$$

Using $r\dot{r} = L e \sin v / (GM\mu^2/L) \cdot (\dots)$, i.e. converting $\mu r \dot{r}$ to L times a function of v via $r = \lambda/(1 + e \cos v)$, $\dot{r} = (Le/(\mu\lambda)) \sin v$ (from §16's kinematics), gives the compact form

$$\dot{\vec{A}} = L \left[\vec{F} \times \hat{L} + \frac{e \sin v}{1 + e \cos v} \left(\frac{(\dot{\vec{r}} \cdot \vec{F})}{\dot{r}} \hat{r} - \vec{F} \right) \right]. \quad (61)$$

14.3 Special case: $\vec{F}$ in the orbital plane

Write $\vec{F} = F_r \hat{r} + F_\phi \hat{\phi}$ (no out-of-plane component). Then:

$$\dot{E} = \dot{r} F_r + r \dot{\phi} F_\phi, \quad (62)$$

$$\dot{L} = r F_\phi, \quad \dot{\hat{L}} = 0. \quad (63)$$

Since $\dot{\hat{L}} = 0$, **the orbital plane does not precess**: $\dot{\Omega} = \dot{i} = 0$ whenever the perturbation is confined to the orbital plane (no out-of-plane component) — this generalizes the exact statement we found for the purely radial example (there, $F_\phi = 0$ too, but the plane-fixing only needed $F_z = 0$).

Working through (61) with $\vec{F} = F_r \hat{r} + F_\phi \hat{\phi}$ (using $\hat{r} \times \hat{L} = -\hat{\phi}$, $\hat{\phi} \times \hat{L} = \hat{r}$, and $r \dot{\phi} / \dot{r} = (1 + e \cos v) / (e \sin v) \equiv 1/\xi$ with $\xi \equiv e \sin v / (1 + e \cos v)$) gives, after simplification,

$$\dot{\vec{A}} = L \left[2F_\phi \hat{r} - (F_r + \xi F_\phi) \hat{\phi} \right]. \quad (64)$$

Rate of $|\vec{A}|$ (i.e. $\dot{e}$). Projecting (64) onto $\hat{A} = \cos \omega \hat{\phi} + \sin \omega \hat{y}$ (using $\hat{r} \cdot \hat{A} = \cos v$, $\hat{\phi} \cdot \hat{A} = -\sin v$, since $v = \phi - \omega$):

$$\dot{A} \equiv \dot{\vec{A}} \cdot \hat{A} = L \left[F_r \sin v + F_\phi (2 \cos v + \xi \sin v) \right]. \quad (65)$$

(Since $A = GM\mu^2 e$, this directly gives $\dot{e} = \dot{A} / (GM\mu^2)$.)

Rate of $\hat{A}$'s rotation (i.e. $\dot{\omega}$). Subtracting off the magnitude-changing piece, $A\dot{\hat{A}} = \dot{\vec{A}} - \dot{A}\hat{A}$, and using $\hat{A} = \cos v \hat{r} - \sin v \hat{\phi}$ (which follows from $\hat{A} \cdot \hat{r} = \cos v$, $\hat{A} \cdot \hat{\phi} = -\sin v$ together with orthonormality) one finds, after simplification,

$$A\dot{\hat{A}} = -L \left[F_r \cos v + F_\phi (\xi \cos v - 2 \sin v) \right] (\sin v \hat{r} + \cos v \hat{\phi}). \quad (66)$$

Since $\hat{A}$ rotates purely by ω changing (with Ω, ι fixed), $\dot{\hat{A}}$ has magnitude $|\dot{\omega}|$ in the direction transverse to $\hat{A}$; carrying through the projection (this is the step where the reference notes work through dotting with $\hat{\phi}$ and dividing by $\sin \omega$ — algebra that is fiddly but standard) gives

$$\dot{\omega} = -\frac{1}{e} \sqrt{\frac{a(1-e^2)}{GM\mu^2}} \left[F_r \cos v + F_\phi (\xi \cos v - 2 \sin v) \right]. \quad (67)$$

Rate of a . From $E = -GM\mu/(2a) \Rightarrow \dot{a} = (2a^2/GM\mu)\dot{E}$, and (62) with $\dot{r} = (Le/\mu\lambda) \sin v$, $r\dot{\phi} = L/(\mu r) = (L/\mu\lambda)(1 + e \cos v)$:

$$\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}} \left[e \sin v F_r + (1 + e \cos v) F_\phi \right]. \quad (68)$$

Rate of e . From (65) and $A = GM\mu^2 e$:

$$\dot{e} = \sqrt{\frac{a(1-e^2)}{GM\mu^2}} \left[F_r \sin v + F_\phi (2 \cos v + \xi \sin v) \right]. \quad (69)$$

Rate of true anomaly. From $\cos v = \hat{A} \cdot \hat{r}$ differentiated, and using $\dot{\phi} = L/(\mu r^2)$:

$$\dot{v} = \sqrt{\frac{GM}{a^3(1-e^2)^3}} (1 + e \cos v)^2 - \dot{\omega}. \quad (70)$$

Osculating-element summary (planar perturbation).

$$\begin{aligned} \dot{a} &= \sqrt{\frac{4a^3}{GM\mu^2(1-e^2)}} \left[e \sin v F_r + (1 + e \cos v) F_\phi \right], \\ \dot{e} &= \sqrt{\frac{a(1-e^2)}{GM\mu^2}} \left[F_r \sin v + F_\phi (2 \cos v + \xi \sin v) \right], \\ \dot{\Omega} &= 0, \quad i = 0, \\ \dot{\omega} &= -\frac{1}{e} \sqrt{\frac{a(1-e^2)}{GM\mu^2}} \left[F_r \cos v + F_\phi (\xi \cos v - 2 \sin v) \right], \\ \dot{v} &= \sqrt{\frac{GM}{a^3(1-e^2)^3}} (1 + e \cos v)^2 - \dot{\omega}. \end{aligned}$$

14.4 Nearly circular orbits: Laplace–Lagrange variables

As $e \rightarrow 0$, ω becomes ill-defined (periapsis is not a meaningful point on a circle) and $\dot{\omega}$ formally diverges in (67). The fix is to use the Cartesian components of $\vec{A}$ directly instead of its polar decomposition:

$$\epsilon_1 \equiv e \cos \omega = \frac{A_1}{GM\mu^2}, \quad \epsilon_2 \equiv e \sin \omega = \frac{A_2}{GM\mu^2}, \quad (71)$$

together with $\Phi \equiv \omega + \ell = n(t - T)$ (T = time of ascending-node passage) in place of v . To $O(e)$, the orbit parametrization becomes

$$r = a(1 - \epsilon_1 \cos \Phi - \epsilon_2 \sin \Phi), \quad (72)$$

$$\phi = \Phi + 2\epsilon_1 \sin \Phi - 2\epsilon_2 \cos \Phi, \quad (73)$$

$$\dot{r} = \frac{L}{a\mu} (\epsilon_1 \sin \Phi - \epsilon_2 \cos \Phi), \quad (74)$$

$$\dot{\phi} = \frac{L}{a^2\mu} (1 + 2\epsilon_1 \cos \Phi + 2\epsilon_2 \sin \Phi). \quad (75)$$

Repeating the perturbation analysis with $\delta\phi \equiv \phi - \Phi = \epsilon_1 \sin \Phi - \epsilon_2 \cos \Phi$ (to leading order in e) gives the osculating equations in this basis:

$$\dot{a} = \sqrt{\frac{4a^3}{GM\mu^2}} \left[\delta\phi F_r + (1 + \epsilon_1 \cos \Phi + \epsilon_2 \sin \Phi) F_\phi \right], \quad (76)$$

$$\dot{\epsilon}_1 = \sqrt{\frac{a}{GM\mu^2}} \left[\sin \Phi F_r + 2 \cos \Phi F_\phi + \delta\phi (2 \cos \Phi F_r - 3 \sin \Phi F_\phi) \right], \quad (77)$$

$$\dot{\epsilon}_2 = \sqrt{\frac{a}{GM\mu^2}} \left[-\cos \Phi F_r + 2 \sin \Phi F_\phi + \delta\phi (2 \sin \Phi F_r + 3 \cos \Phi F_\phi) \right], \quad (78)$$

$$\dot{i} = 0, \quad \dot{\Omega} = 0, \quad \dot{\Phi} = \sqrt{\frac{GM}{a^3}} + O(e^2). \quad (79)$$

Using Φ as the independent variable ($dt/d\Phi = \sqrt{a^3/GM}$) removes explicit time dependence:

$$\frac{da}{d\Phi} = \frac{2a^3}{GM\mu} \left[\delta\phi F_r + (1 + \epsilon_1 \cos \Phi + \epsilon_2 \sin \Phi) F_\phi \right], \quad (80)$$

$$\frac{d\epsilon_1}{d\Phi} = \frac{a^2}{GM\mu} \left[\sin \Phi F_r + 2 \cos \Phi F_\phi + \delta\phi (2 \cos \Phi F_r - 3 \sin \Phi F_\phi) \right], \quad (81)$$

$$\frac{d\epsilon_2}{d\Phi} = \frac{a^2}{GM\mu} \left[-\cos \Phi F_r + 2 \sin \Phi F_\phi + \delta\phi (2 \sin \Phi F_r + 3 \cos \Phi F_\phi) \right]. \quad (82)$$

15 Rømer Delay for a Keplerian Orbit

For a signal emitted from the primary body, the light-travel-time (Rømer) delay from the changing distance to the barycentre is $\Delta_R = -\frac{1}{c} \vec{x}_1 \cdot \hat{R} = -\frac{1}{c} \frac{m_2}{M} \vec{r} \cdot \hat{R}$. Using $\vec{r} = r(v) (\cos \phi \hat{\delta} + \sin \phi \hat{\gamma})$ and the definitions (5)–(6) of $\hat{\delta}, \hat{\gamma}$ in terms of $\hat{\alpha}, \hat{\delta}, \hat{R}$, all terms proportional to $\hat{\alpha} \cdot \hat{R}, \hat{\delta} \cdot \hat{R}$ (which vanish by orthogonality) drop, leaving only the $\hat{R} \cdot \hat{R} = 1$ piece:

$$\Delta_R = \frac{1}{c} \frac{m_2}{M} r(v) \sin \phi \sin \iota = \frac{1}{c} \frac{m_2}{M} r(v) \sin(v + \omega) \sin \iota.$$

Expanding $\sin(v + \omega) = \cos v \sin \omega + \sin v \cos \omega$ and substituting $r \cos v = a(\cos u - e)$, $r \sin v = a\sqrt{1 - e^2} \sin u$ (from (31) and $r = a(1 - e \cos u)$):

$$\Delta_R = \frac{a_1 \sin \iota}{c} \left[(\cos u - e) \sin \omega + \sqrt{1 - e^2} \sin u \cos \omega \right], \quad a_1 \equiv \frac{m_2}{M} a. \quad (83)$$

16 Worked Example: the Radial Perturbation $\vec{F} = \alpha \hat{r}/r^3$

This section reproduces, with every intermediate step shown, the fully hand-derived example from our earlier session, then cross-checks each result against the general machinery of §14.1–15. Take a purely radial perturbing force

$$\vec{F} = \frac{\alpha}{r^3} \hat{r}, \quad (84)$$

equivalently derivable from a perturbing potential energy $V_{\text{pert}}(r) = \alpha/(2r^2)$, since $\vec{F} = -\nabla V_{\text{pert}} = -\frac{dV_{\text{pert}}}{dr} \hat{r} = \frac{\alpha}{r^3} \hat{r}$. This is the classic exactly-solvable "precessing ellipse" problem (an inverse-cube force added to Newtonian gravity).

16.1 Energy rate and $\dot{a}$

Dot the perturbed equation of motion (54) with $\dot{\vec{r}}$:

$$\mu \ddot{\vec{r}} \cdot \dot{\vec{r}} = -\frac{GM\mu}{r^2}(\hat{r} \cdot \dot{\vec{r}}) + \vec{F} \cdot \dot{\vec{r}}.$$

Now $\hat{r} \cdot \dot{\vec{r}} = \dot{r}$ (radial speed), $\mu \ddot{\vec{r}} \cdot \dot{\vec{r}} = \frac{d}{dt}(\frac{1}{2}\mu \dot{\vec{r}}^2) = \dot{T}$, and with $V(r) = -GM\mu/r$, $\dot{V} = (GM\mu/r^2)\dot{r}$, so $-GM\mu\dot{r}/r^2 = -\dot{V}$. Hence $\dot{T} = -\dot{V} + \vec{F} \cdot \dot{\vec{r}}$, i.e.

$$\dot{E} = \vec{F} \cdot \dot{\vec{r}} \quad (E \equiv T + V), \quad (85)$$

which is just (58) specialized here — only the perturbing force changes the osculating Kepler energy, since gravity is already absorbed into E . From $E = -GM\mu/(2a)$ (Eq. 26), $dE/da = GM\mu/(2a^2)$, so

$$\frac{GM\mu}{2a^2}\dot{a} = \vec{F} \cdot \dot{\vec{r}} \implies \dot{a} = \frac{2a^2}{GM\mu}\vec{F} \cdot \dot{\vec{r}}.$$

For (84), $\vec{F} \cdot \dot{\vec{r}} = (\alpha/r^3)(\hat{r} \cdot \dot{\vec{r}}) = (\alpha/r^3)\dot{r}$, so

$$\dot{a} = \frac{2\alpha a^2}{GM\mu r^3} \dot{r}. \quad (86)$$

16.2 Assembling $\dot{a}(u)$

Using the eccentric-anomaly kinematics already established in (30) and (39) ($r = a(1 - e \cos u)$, $\dot{r} = a n e \sin u / (1 - e \cos u)$) and $r^3 = a^3(1 - e \cos u)^3$:

$$\dot{a} = \frac{2\alpha e n \sin u}{GM\mu(1 - e \cos u)^4}. \quad (87)$$

16.3 No secular drift in a

$\sin u$ is odd about $u = 0, \pi$ while $(1 - e \cos u)^{-4}$ is even, so $\dot{a}$ integrates to zero over a full period. Converting $dt = (1 - e \cos u) du/n$ (from differentiating Kepler's equation 34) and integrating (87) over $u : 0 \rightarrow 2\pi$ (verified symbolically with `sympy` in the earlier session) gives

$$\Delta a_{\text{orbit}} \equiv \oint \dot{a} dt = 0 \quad \text{exactly.} \quad (88)$$

So a oscillates within an orbit but returns to its starting value — no secular drift. This is consistent with $\vec{F}$ being conservative: the *true* energy of the full system, $E_{\text{true}} = T + V + V_{\text{pert}}$, is exactly conserved at every instant (it is an autonomous conservative system), whereas $E = -GM\mu/2a$ above is only the *osculating* Kepler energy (what the energy would be if the perturbation switched off right now). The oscillation of a tracks the difference between these two, not a violation of energy conservation.

16.4 Angular momentum: exact conservation

From $\dot{\vec{L}} = \vec{r} \times \vec{F}$ (Eq. 59), and since $\vec{F} = (\alpha/r^3)\hat{r} \parallel \vec{r}$ here,

$$\dot{\vec{L}} = \vec{r} \times \vec{F} = 0 \quad \text{exactly (pointwise, not merely on average).} \quad (89)$$

This is stronger than the general planar-perturbation statement $\dot{\vec{L}} = 0$ of §15.2 (which only requires the *sum* of torque contributions to vanish for $F_\phi \neq 0$ but no out-of-plane component); here $\vec{r} \times \vec{F}$ vanishes term-by-term because $\vec{F}$ is exactly radial. Consequently ι, Ω are exactly constant — the orbital plane never moves.

16.5 $\dot{e}$ from $L^2 = \text{const}$

From $L^2 = GM\mu^2 a(1 - e^2)$ (Eq. 28) and $\dot{\vec{L}} = 0 \Rightarrow (\dot{L}^2) = 0$ exactly:

$$0 = GM\mu^2 \left[(1 - e^2)\dot{a} - 2ae\dot{e} \right] \implies \dot{e} = \frac{1 - e^2}{2ae}\dot{a}.$$

This holds pointwise; since $\langle \dot{a} \rangle = 0$ over an orbit (88), $\langle \dot{e} \rangle = 0$ too.

16.6 The Laplace–Runge–Lenz vector: general perturbed rate

We now rederive $\dot{\vec{A}} = \vec{F} \times \vec{L}$ (already stated as the reduction of Eq. 60 in §15.1) with every step shown, since it is the key tool for the precession rate. Starting from $\vec{A} = \vec{p} \times \vec{L} - GM\mu^2 \hat{r}$:

$$\dot{\vec{A}} = \dot{\vec{p}} \times \vec{L} + \vec{p} \times \dot{\vec{L}} - GM\mu^2 \dot{\hat{r}}.$$

Here $\dot{\vec{L}} = 0$ exactly (89), so the middle term drops. With $\dot{\vec{p}} = \mu \ddot{\vec{r}} = -\frac{GM\mu}{r^2} \hat{r} + \vec{F}$ (Newton's second law):

$$\dot{\vec{A}} = -\frac{GM\mu}{r^2} (\hat{r} \times \vec{L}) + \vec{F} \times \vec{L} - GM\mu^2 \dot{\hat{r}}.$$

Key cancellation. For planar motion with $\vec{L} = L\hat{L}$, write $\hat{r} = \cos\phi \hat{x} + \sin\phi \hat{y}$, $\hat{\phi} = -\sin\phi \hat{x} + \cos\phi \hat{y}$ (a local Cartesian frame with $\hat{x} \times \hat{y} = \hat{L}$). Direct computation gives $\hat{r} \times \hat{L} = -\hat{\phi}$, and differentiating $\hat{r}$: $\dot{\hat{r}} = \dot{\phi} \hat{\phi}$. Using $L = \mu r^2 \dot{\phi}$ (Eq. 17), so $\dot{\phi} = L/(\mu r^2)$:

$$\begin{aligned} -\frac{GM\mu}{r^2} (\hat{r} \times \vec{L}) &= -\frac{GM\mu}{r^2} \cdot L(\hat{r} \times \hat{L}) = -\frac{GM\mu L}{r^2} (-\hat{\phi}) = \frac{GM\mu L}{r^2} \hat{\phi}, \\ GM\mu^2 \dot{\hat{r}} &= GM\mu^2 \dot{\phi} \hat{\phi} = GM\mu^2 \cdot \frac{L}{\mu r^2} \hat{\phi} = \frac{GM\mu L}{r^2} \hat{\phi}. \end{aligned}$$

These two terms are identical, so they cancel exactly:

$$\boxed{\dot{\vec{A}} = \vec{F} \times \vec{L}} \tag{90}$$

— the same general result as Eq. (60) reduced, but now with the cancellation made completely explicit (this is exactly why $\vec{A}$ is conserved for the pure Kepler problem, $\vec{F} = 0$).

16.7 Specializing to $\vec{F} = \alpha \hat{r}/r^3$

$$\dot{\vec{A}} = \frac{\alpha}{r^3} \hat{r} \times L\hat{L} = \frac{\alpha L}{r^3} (\hat{r} \times \hat{L}) = -\frac{\alpha L}{r^3} \hat{\phi}.$$

16.8 Extracting $\dot{\omega}$

$\vec{A}$ points from the focus towards periapsis: $\vec{A} = A(\cos\omega \hat{x} + \sin\omega \hat{y})$ (identifying $\hat{x} = \hat{\mathcal{O}}$; cf. Eq. 53). Its own comoving tangential direction is $\hat{\phi}_\omega \equiv -\sin\omega \hat{x} + \cos\omega \hat{y}$, and

$$\dot{\vec{A}} = \dot{A} \hat{A} + A \dot{\omega} \hat{\phi}_\omega.$$

Projecting (16.7) onto $\hat{\phi}_\omega$ (using $\hat{\phi}_\omega \cdot \hat{A} = 0$ and $\hat{\phi}_\omega \cdot \hat{\phi} = \cos(\phi - \omega) = \cos v$, since the angle between two azimuthal unit vectors equals the angle between their base directions, and $v \equiv \phi - \omega$ is exactly the true anomaly):

$$A \dot{\omega} = \hat{\phi}_\omega \cdot \dot{\vec{A}} = -\frac{\alpha L}{r^3} \cos v \implies \dot{\omega} = -\frac{\alpha L \cos v}{A r^3}.$$

16.9 Orbit-averaged precession rate

Using $r = a(1 - e^2)/(1 + e \cos v)$, $dt = \mu r^2 dv/L$ (from $\dot{\phi} = \dot{v} = L/\mu r^2$), $A = GM\mu^2 e$ (Eq. 52), $L = \mu\sqrt{GMa(1 - e^2)}$ (Eq. 28), and $T = 2\pi\sqrt{a^3/GM}$ (Eq. 33), integrate (16.8) over $v : 0 \rightarrow 2\pi$. Carrying this out (done symbolically with `sympy` in the earlier session) gives

$$\Delta\omega_{\text{orbit}} = -\frac{\pi\alpha}{GM\mu a(1 - e^2)}, \quad \langle\dot{\omega}\rangle = -\frac{\alpha}{2\mu\sqrt{GM} a^{5/2}(1 - e^2)}. \quad (91)$$

So while a, e merely oscillate with zero net drift (88, 16.5), ω precesses secularly at this rate — the apsidal precession the perturbation produces.

Element	General planar perturbation (F_r, F_ϕ)	Purely radial α/r^3 perturbation
ι, Ω	Constant to the extent $\hat{L}$ is fixed (exact if $F_z = 0$)	Exactly constant
a	Eq. (68)	Oscillates, zero secular drift
e	Eq. (69)	Oscillates, zero secular drift
ω	Eq. (67)	Secular precession, Eq. (91)